

ABHIUDAYA PRATAP SINGH

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Education

B.Tech in CSE(Artificial Intelligence and Machine learning)

Vellore Institute of Technology, Bhopal

2023 – 2027

CGPA: 8.60/10

Senior Secondary (12th)

Army Public School, Lucknow

2023

Percentage: 76.5 %

Secondary School (10th)

Army Public School, Lucknow

2021

Percentage: 90.0%

Technical Skills

- **Programming Languages:** Java, Python
- **Generative AI & LLMs:** Retrieval-Augmented Generation (RAG), Large Language Model (LLM) Fundamentals, Prompt Engineering, Conversational AI, Google Gemini API
- **Machine Learning & AI:** Machine Learning, Scikit-learn, Pandas, NumPy
- **Backend & APIs:** Node.js, Express.js, REST APIs, JWT Authentication, API Integration
- **Frontend Development:** React, HTML, CSS
- **Databases & Tools:** MongoDB, MySQL, DBMS, GitHub

Projects

Smart Health AI (AI-Powered Healthcare Platform)

Jul 2026

Tech Stack: React.js, Node.js, Express.js, MongoDB, Mongoose, JWT, Google Gemini API

[GitHub](#) — [Live Demo](#)

- Designed and developed a full-stack healthcare platform that delivers AI-powered health assistance through real-time conversational interactions powered by LLMs.
- Integrated Google Gemini API to deliver intelligent, conversational responses to health-related queries, demonstrating LLM integration and understanding of API consumption.
- Built a scalable REST API backend with Node.js and Express.js, implementing JWT-based authentication for secure user management and API interactions.
- Implemented a responsive React.js frontend with intuitive UI/UX design for seamless user interaction with AI agents.
- Utilized MongoDB with Mongoose for efficient user data storage and conversation history management.

Diabetes Prediction Machine Learning System

Jan 2024

Tech Stack: Python, Scikit-learn, Pandas, NumPy, HTML, CSS, Streamlit

[GitHub](#) — [Live Demo](#)

- Developed a machine learning system to assess health risk using key medical parameters and real-world data.
- Implemented Logistic Regression classification and optimized performance using precision and F1-score metrics.
- Applied data preprocessing, feature selection, and model validation to improve prediction reliability and accuracy.
- Built and deployed an interactive web interface for instant health risk predictions and customer-facing deployment.

Voice Command Shopping Assistant (VoiceCart)

Aug 2024

Tech Stack: React, Node.js, Express.js, TypeScript, Tailwind CSS, Web Speech API

[GitHub](#)

- Engineered a full-stack voice-interactive shopping application that processes natural language voice commands and converts them into structured intents and entities using rule-based NLP algorithms.
- Implemented a command parser that extracts intent, quantities, product names, and brands from user voice input, demonstrating proficiency in conversational AI and natural language understanding.
- Developed a responsive React frontend with Web Speech API integration for real-time voice recognition, enabling intuitive voice-to-text interaction patterns.
- Built a RESTful Express.js backend API with product management endpoints and CORS configuration, handling shopping list operations and product search functionality with mock database integration.

Leadership and Experience

- **PR and Outreach Member, iCreate Club:** Managed partnerships and coordinated technical events to enhance club visibility. Demonstrated customer-oriented approach by working directly with industry professionals and translating business needs into collaborative solutions.
- **Experiential Internship – Industry Exposure:** Visited tech companies including L&T, Zensor Technologies, BSNL and others. Gained practical insights into industry operations, real-world software development practices, problem-solving methodologies, and networking with tech professionals.

Certifications

- Agentic AI Certified Foundations Associate, Oracle (Aug 2026)
- Applied Machine Learning in Python, Coursera (Dec 2025)
- Deloitte Australia – Data Analytics Job Simulation, Forage (Jul 2025)
- Cloud Computing, NPTEL (May 2025)